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DATABASES ASSIGNMENT Date: _____

Q1) 1) $\pi_{country_name} (\sigma_{security_level = 'Critical'} (THREAT) \bowtie_{THREAT_Threat_ID = INCIDENT_Threat_ID} INCIDENT)$

$\bowtie_{INCIDENT_Country_ID = COUNTRY_Country_ID} COUNTRY$
 $\sigma_{Date_reported \text{ BETWEEN } '2025-01-01' \text{ AND } '2025-12-31'}$

First pick all threats marked as Critical, link them to incidents that happened in 2025, then match incidents with their country and display country's name.

2) $\pi_{Team_name} (\sigma_{Threat_Name = 'Ransomware'} (THREAT) \bowtie_{THREAT_Threat_ID = INCIDENT_Threat_ID} INCIDENT)$

$\bowtie_{INCIDENT_Incident_ID = ACTION_TAKEN_Incident_ID}$
 $\sigma_{Resolution_Status = 'Resolved'} (ACTION_TAKEN)$

$\bowtie_{ACTION_TAKEN_Team_ID = RESPONSE_TEAM_Team_ID} RESPONSE_TEAM$

Find incidents caused by 'ransomware', filter actions with 'resolved status', join to RESPONSE_TEAM and team names.

3) $\pi_{country_name} (COUNTRY \bowtie_{COUNTRY_Country_ID = RESPONSE_TEAM_Country_ID} RESPONSE_TEAM)$

Subtract countries that appear in response teams from all countries.

Date: _____

4) $T_{count} \leftarrow \gamma_{Threat-ID; Count} \text{ distinct}(Country-ID) \rightarrow \pi_{Threat-ID, Count} (INCIDENT)$
 $\pi_{Threat-ID, Count} (\sigma_{Count > 1} (T_{count})) \bowtie THREAT$

group INCIDENT by Threat-ID country distinct reporting countries
keep Threat-IDs with count > 1, join with threat to get names.

5) $\pi_{Incident-ID, Country-Name, Impact-Score} (\sigma_{Impact-Score > 80} (Category = 'AI Attack' (THREAT) \bowtie THREAT.Threat-ID = INCIDENT.Threat-ID INCIDENT \bowtie INCIDENTa (Country-ID = COUNTRY.Country-ID (COUNTRY)))$

Select incidents where threat is 'AI Attack' and score > 80, join to country, and project incident-ID, country-name and impact-score.

- 1) Select the names of menu items that are not available and cost more than 500.
- 2) Select the names of items that are either a Beverage with price > 300 or a Desert
- 3) Select the name and price of all available snacks
- 4) Select the name and price / 100 of all available snacks
- 5) Find Supplier IDs of suppliers that supply Cappuccinos.
- 6) Show the employee name & cafe name for employees who work at a cafe located in the same city as their address
- 7) List employee IDs of all employees whose salary is not 50,000
- 8) Find supplier IDs that supply both Karachi & Lahore
- 9) List item names that appear on the Menu items table but have never been supplied